

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and quantitative researcher with experience in stochastic simulation, scalable likelihood-based modeling, statistical diagnostics, and financial risk analytics. Builds modular Python/PyTorch research workflows for large, noisy dependent datasets and translates model results into clear, decision-relevant insights. Brings finance-sector experience and a strong interest in capital markets, asset allocation, and systematic investment research.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ Expected Dec 2026 | GPA: 3.8/4.0
M.S. in Statistics, University of Minnesota, Minneapolis, MN Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Research Systems: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), Linux, Git, AWS EC2, HPC/SLURM; reproducible pipelines, cached artifacts, CPU/GPU execution, restartable workflows

Quantitative Research: Stochastic processes, Monte Carlo simulation, time-series and spatiotemporal modeling, Gaussian processes, MLE/GLS, numerical optimization, regression, spectral analysis, predictive modeling, model validation, scenario and sensitivity analysis

EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Quantitative Finance / Risk Modeling Summer 2026

- Developed and validated a Sequential Probability Ratio Test (SPRT) framework for binary risk indicators, modeling the cumulative log-likelihood-ratio process as a random walk with statistically controlled decision boundaries.
- Applied a Brownian-motion approximation and dynamic-programming probability propagation to quantify boundary-crossing probabilities, stopping-time distributions, Type I/II error, and early-decision tradeoffs.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team Summer 2025

- Developed an end-to-end LightGBM pricing pipeline across 2.46M+ records, including data preparation, GLM benchmarking, model validation, and performance diagnostics on AWS EC2.
- Presented model diagnostics and feature-attribution findings as pricing and risk-segmentation recommendations for business stakeholders.

QUANTITATIVE RESEARCH AND MARKETS

Ph.D. Dissertation Research - Scalable Stochastic Modeling and Research Workflows Sep 2024 - Present

- Stochastic Modeling and Simulation:** Developed an advection-aware Vecchia likelihood approximation for nonseparable Gaussian processes, replacing dense covariance operations with fixed-budget conditioning to fit up to 145,008 observations per day.
- Research Diagnostics:** Compared Matérn and generalized Cauchy dependence models and designed scale- and frequency-resolved tests to identify misspecification in smoothness, noise, and dependence.
- Data and Workflow Engineering:** Built modular Python/PyTorch pipelines for quality filtering, missing-data handling, nearest-neighbor construction, dependency-aware ordering, irregular-grid resampling, cached outputs, and restartable HPC execution.
- Estimation and Scalability:** Implemented GLS profiling, L-BFGS optimization, Monte Carlo simulation, and CPU/GPU execution for repeatable large-scale model estimation and evaluation.

IMC Prosperity Algorithmic Trading Competition - Top 2.39% of Participants Apr 2025

- Built and evaluated a market-making strategy with inventory-aware quoting, dynamic liquidation rules, and benchmark-driven performance analysis.

ADDITIONAL

Coursework: Econometrics, Machine Learning, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Microeconomics

Languages: Korean (Native); English (Fluent); Chinese (Basic)